SOFTWARE ENGINEERING · LECTURE 4

Scan to follow along

Open today's slides on your own phone or laptop — then you can run real code yourself.

`hassanmiir.github.io/cpp-course/lecture4/`



📱 Point your camera here

👁 0 opened today

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LECTURE 4 · MAKING DECISIONS

Teach your program to choose

Pass or fail. Login allowed or denied. Discount or full price.

Every app you've ever used is making **thousands of choices a second** — with one keyword: `if`.

A 2-hour hands-on session · you'll build a grade classifier

WHERE WE LEFT OFF

Last time your pass check printed `1` or `0`

```
bool passed = average >= 50;
cout << passed;     // prints 1 or 0 — not very friendly
```



yes → print "PASS"
no → print "FAIL"

A `bool` is a *yes/no*. But the program still did the **same thing** either way. Today we make it **act** on that answer.

PART 1

The if statement

Run some code — but only when a condition is true.

- One condition, one path
- Built from the bool you already know
- Run your first decision — live

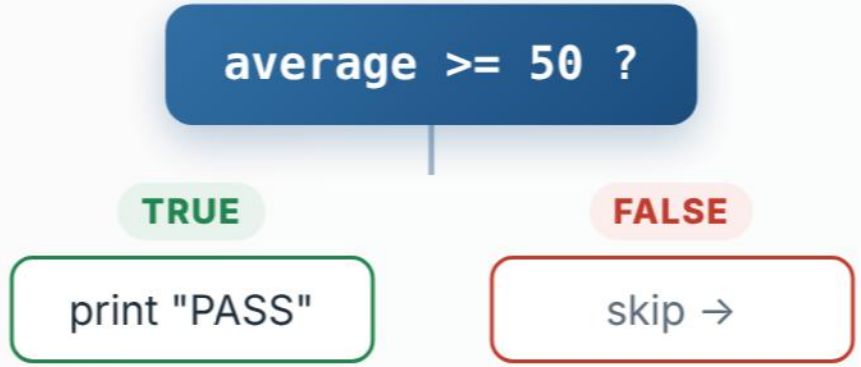


THE SHAPE

A condition, then a block

```cpp
if (average >= 50) {
    cout << "PASS";
}
```

Read it: *"if this is true, run what's in the braces."* If it's false, C++ just skips the block.



YOUR TURN · LIVE CODE RUN IT!

Your first decision

Run it. Then change the mark to 40 and run again — watch the message disappear. (Input: 72)

main.cpp ▶ Run ↺ Open in OnlineGDB ▸

```
#include <iostream>
using namespace std;

int main() {
    int mark;
    cin >> mark;

    if (mark >= 50) {
        cout << "PASS" << endl;
    }

    cout << "Done " << endl;    // this always runs
```

OUTPUT Ready

Click ▶ Run to see the output here.

PART 2

else, and the ladder

Give the "no" path something to do — and chain many choices.

- else — the other path
- else if — a ladder of choices
- Build a grade classifier



TWO PATHS

if … `else` … — always one or the other

```cpp
if (mark >= 50) {
    cout << "PASS";
} else {
    cout << "FAIL";
}
```



Exactly one runs
Never both, never neither. That's the guarantee `else` gives you.

MANY PATHS

The `else if` ladder — checked top to bottom

```
if (score >= 90) {
    grade = 'A';
} else if (score >= 75) {
    grade = 'B';
} else if (score >= 50) {
    grade = 'C';
} else {
    grade = 'F';
}
```

Top to bottom

C++ tries each test in order and stops at the **first** true one.

Order matters!

Put the strictest test first. 95 is ≥90 and ≥75 — but only 'A' runs.

else = catch-all

If nothing matched, the final `else` handles it.

YOU BE THE COMPUTER ON PAPER · 3 MIN

Trace the ladder — which grade?

Run the ladder above **in your head** for each score. Write the letter it lands on.

`score = 95` grade = ?

`score = 75` grade = ?

`score = 49` grade = ?

`score = 50` grade = ?

✏️ **Go line by line. Stop at the FIRST test that's true — that's the trick.**

THE ANSWER

Check your four

`score = 95`	→	`'A'`	95 ≥ 90, stops immediately
`score = 75`	→	`'B'`	not ≥90, but 75 ≥ 75 ✓
`score = 49`	→	`'F'`	fails every test → else
`score = 50`	→	`'C'`	50 ≥ 50 ✓ (the boundary!)

Boundaries bite

Beginners trip on `>=` vs `>`. 50 passes because the test is `>= 50`, not `> 50`.

YOUR TURN · LIVE CODE RUN IT!

The grade classifier

The real tool Group B's gradebook needs. Run it, then feed it 95, 60, 30. (Input: 72)

main.cpp ▶ Run ↺ Open in OnlineGDB ▸

```
#include <iostream>
using namespace std;

int main() {
    int score;
    cin >> score;
    char grade;

    if (score >= 90)      grade = 'A';
    else if (score >= 75) grade = 'B';
    else if (score >= 50) grade = 'C';
    else                  grade = 'F';
```

OUTPUT Ready

Click ▶ Run to see the output here.

QUICK CHECK THINK!

What does this print for `score = 95`?

```
if (score >= 50)      cout << "C";
else if (score >= 75) cout << "B";
else if (score >= 90) cout << "A";
```

Careful — look at the *order* of the tests.

A C

B A

C B

PART 3

switch — the menu

A clean way to pick one of many fixed options.



One value, many cases

Don't forget `break` !

THE MENU

`switch` — pick a case by value

```
switch (grade) {
  case 'A': cout << "Excellent"; break;
  case 'B': cout << "Good";      break;
  case 'C': cout << "Passed";    break;
  default:  cout << "Try again";
}
```

When to use it

One variable, compared to several **exact** values — menus, grades, commands.

The `break` trap

Forget `break` and it "falls through" into the next case. The #1 switch bug.

`default`

The catch-all, like the final `else`.

QUICK CHECK THINK!

No `break`! What prints for `'A'`?

```
switch (grade) {
  case 'A': cout << "Excellent";
  case 'B': cout << "Good";
  default:  cout << "!";
}
```

A Excellent

B ExcellentGood!

C !

PART 4

Build it: smart grade report

Turn a score into a grade AND a message — your real classifier.

- Score → letter grade
- A message for each grade
- Work with a partner



PAIR CHALLENGE · USE YOUR BRAIN 15 MIN

The smart grade report

Read a score and print a full report:

1. Use an `if / else if` ladder to set a letter grade (A/B/C/F).
2. Then use a `switch` on that grade to print a message.
3. Guard it: if the score is below 0 or above 100, print `"Invalid score"` and nothing else.
4. Print: the score, the grade, and the message.

15:00

❚❚ Pause ↺ +1 min

Don't forget

A `break` after every `case`, and check the valid range *first* with an `if`.

PAIR CHALLENGE · BUILD HERE RUN IT!

Your grade report

Starter reads the score & sets the grade. **You add** the switch message and the range guard. (Input: 82)

main.cpp ▶ Run ↺ Open in OnlineGDB ▸

```
#include <iostream>
using namespace std;

int main() {
    int score;
    cin >> score;

    // TODO: if (score < 0 || score > 100) { print "Invalid score"; return 0; }

    char grade;
    if (score >= 90)      grade = 'A';
    else if (score >= 75) grade = 'B';
```

OUTPUT Ready

Click ▶ Run to see the output here.

ONE SOLUTION COMPARE YOURS

How the report works

```cpp
if (score < 0 || score > 100) {
    cout << "Invalid score"; return 0;   // guard FIRST
}
char grade;
if (score >= 90) grade = 'A';
else if (score >= 75) grade = 'B';
else if (score >= 50) grade = 'C';
else grade = 'F';

cout << "Score: " << score << "  Grade: " << grade << endl;
switch (grade) {
  case 'A': cout << "Outstanding!"; break;
  case 'B': cout << "Well done";    break;
  case 'C': cout << "You passed";   break;
  default:  cout << "Keep going";
}
```

Guard first, return early

Reject bad input before any grading runs.

Ladder sets the grade

Strictest test on top, catch-all `else` at the bottom.

switch speaks to the user

One message per grade — every `case` has its `break`.

RECAP

What you can do now

Branch

`if` / `else` / `else if` — run different code for different data, checked top to bottom.

Switch

Pick one of many fixed values cleanly — with a `break` after each case.

Milestone

Your programs now **react** — they take different paths for different input. That's real logic.

KEEP THESE WORDS

Today's keywords — click any one for a reminder

`if` Run if true	**`else`** Otherwise	**`else if`** Next test	**`switch`** Pick a case
`break` Stop the case	**`default`** Catch-all	**`condition`** true / false test	**`boundary`** The edge case

BEFORE NEXT TIME

Practice (15–20 min)

Leap year checker

Read a year; print "Leap" if divisible by 4 (and your rule) — great `if` + `%` practice.

Day of the week

Read a number 1–7; use a `switch` to print the day name (default: "Invalid").

Upgrade your classifier

Add A+ and A- bands to your grade report — Group B will reuse it.

NEXT LECTURE

Lecture 5 — Loops

Today you chose a path *once*. Next time you'll **repeat** work — grade a whole class of students, not just one, without copying code.



- `while` and `for` — do it again and again
- Loop over every student's score
- Sum, average, and find the top mark



Your program can decide.

From a yes/no to real branching paths — that's the leap from a calculator to a program. Next: doing it again and again.

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